

KU YEE SIANG

AI AUTOMATION & AGENT ENGINEER

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PROFESSIONAL PROFILE

Artificial Intelligence graduate focused on AI automation, agentic systems, and end-to-end product delivery. Built secure multi-agent workflows, LLM evaluation benchmarks, retrieval-augmented applications, and full-stack business tools using Python and TypeScript. Skilled at translating requirements into working solutions, integrating APIs and databases, and improving reliability through validation and safety guardrails.

EDUCATION & AWARD

Beijing Institute of Technology | Bachelor of Science in Artificial Intelligence | 2022-2026

GPA: 3.6/4.0 | Best Student Award

SELECTED PROJECTS

Harness Agent - Enterprise Procurement Workbench | Agent Engineering

- Built an AI-powered procurement workbench that automates product sourcing, supplier comparison, inventory analysis, purchase-order workflows, product onboarding, and report generation.
- Engineered the Vue 3 and FastAPI platform with DeepSeek, MCP/ERP tools, specialist sub-agents, SSE execution streaming, human approval, long-term memory, progressive skills, and per-user OpenSandbox isolation.

Medical LLM Causal Consistency Evaluation Benchmark | Graduation Thesis

- Designed and built MedCausalBenchmark from six NCCN clinical guidelines using a dual-axis taxonomy covering medical scenarios and Basic, Joint, Nested, and Conditional counterfactual reasoning.
- Created a repeatable evaluation workflow comparing Pure LLM, RAG, and CausalSCM across Qwen, LLaMA, and DeepSeek; CausalSCM achieved 91.39% average accuracy.

Mini-Map - Gemini-Powered Virtual Travel Journal | Google Cloud Rapid Agent Hackathon

- Developed a choice-driven travel application with Gemini Pro, Google Cloud Agent Builder, FastAPI, Next.js, and MongoDB Atlas, turning user preferences into journeys constrained by geography and budget.
- Integrated validated MCP tools, GeoJSON persistence, Atlas Vector Search with 1024-dimensional embeddings, external API grounding, deduplication safeguards, and one-click itinerary export.

Agentic Medical Chat System | Full-Stack AI

- Built an end-to-end medical Q&A application combining tool-augmented LLM agents, retrieval, a web interface, and safety guardrails for structured clinical conversations.
- Integrated frontend, agent orchestration, backend retrieval, validation, and error handling into a traceable full-stack workflow for safer medical information delivery.

TECHNICAL SKILLS

Programming

Python, TypeScript, JavaScript, SQL, HTML/CSS

Agentic AI & LLM

Agent orchestration, multi-agent systems, LLM evaluation, benchmark design, RAG, prompt and context engineering, tool calling, human-in-the-loop, memory, safety guardrails

Models & Platforms

DeepSeek, Gemini Pro, Qwen, LLaMA, OpenAI Codex, Google Cloud Agent Builder, Vertex AI, MCP, OpenSandbox

Full-Stack

FastAPI, React, Next.js, Vue 3, Tailwind CSS, REST APIs, Server-Sent Events, API integration

Data & Search

MongoDB Atlas, Atlas Search, Vector Search, SQLite, embeddings, GeoJSON

Engineering

Requirements analysis, debugging, testing, validation, access control, data isolation, Git/GitHub, Vercel

Practices